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LOC Page

Contents

CHAPTER 1 ■ A Resistome-Driven Ensemble Framework for Antimicrobial Resistance Phenotype Prediction	1
JAMIL AS-AD, SAFAET JAMAN ARMAN, KAZI INJAMAMUL HOQUE , and Md. KHALILUZZAMAN	
1.1 INTRODUCTION	2
1.2 LITERATURE REVIEW AND THEORETICAL BACKGROUND	3
1.2.1 Genomic Features for AMR Prediction	3
1.2.2 Machine Learning Models and Ensembles	4
1.3 METHODOLOGY	5
1.3.1 Data Acquisition	5
1.3.2 Data Preprocessing	6
1.3.3 Feature Engineering: Resistome Burden Index	6
1.3.4 Classifier Ensemble Model	8
1.3.5 Experimental Setup	9
1.4 RESULTS AND DISCUSSION	9
1.4.1 Performance of Baseline Models	9
1.4.2 Proposed Ensemble vs. Top-5 Baselines	10
1.4.3 Impact of the Resistome Burden Index	12
1.4.4 Ablation Study	12
1.4.5 Error Analysis	13
1.4.6 Comparison with Existing Methods	14
1.5 CONCLUSION	16



A Resistome-Driven Ensemble Framework for Antimicrobial Resistance Phenotype Prediction

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CONTENTS

1.1	Introduction	2
1.2	Literature Review and Theoretical Background	3
1.2.1	Genomic Features for AMR Prediction	3
1.2.2	Machine Learning Models and Ensembles	4
1.3	Methodology	5
1.3.1	Data Acquisition	5
1.3.2	Data Preprocessing	6
1.3.3	Feature Engineering: Resistome Burden Index	6
1.3.4	Classifier Ensemble Model	8
1.3.5	Experimental Setup	9
1.4	Results and Discussion	9
1.4.1	Performance of Baseline Models	9
1.4.2	Proposed Ensemble vs. Top-5 Baselines	10
1.4.3	Impact of the Resistome Burden Index	11

1.4.4	Ablation Study	12
1.4.5	Error Analysis	13
1.4.6	Comparison with Existing Methods	14
1.5	Conclusion	16

Antimicrobial resistance (AMR) poses a critical global health threat, necessitating rapid and accurate resistance prediction methods. Many current machine learning approaches for genomic AMR prediction rely on binary gene presence/absence features, implicitly treating resistance determinants as independent predictors. This paper introduces a framework that combines two key contributions: the Resistome Burden Index (RBI), a cumulative feature that quantifies the aggregate load of resistance determinants per isolate, and a heterogeneous SVM–XGBoost ensemble classifier. Evaluated across twelve antibiotic–pathogen datasets from the NCBI Pathogen Detection project, the proposed method achieves an average F1-score of approximately 0.94 with recall around 0.95. Compared with previously reported resistome-based approaches, the framework improves F1-score in 8 out of 12 datasets, with a mean improvement of approximately 5% and a maximum improvement of up to 13.8% for specific pathogen–antibiotic combinations. The ensemble’s high recall enables reliable detection of resistant isolates, which is critical for clinical decision-making. To the best of our knowledge, this work represents the first application of a cumulative resistome feature in conjunction with an SVM–XGBoost ensemble for genomic AMR prediction, offering improved predictive performance and computational efficiency over conventional approaches.

Keywords: antimicrobial resistance, machine learning, ensemble methods, SVM, XGBoost, resistome, feature engineering

1.1 INTRODUCTION

Antimicrobial resistance (AMR) represents one of the most pressing global health challenges, with drug-resistant infections causing an estimated 1.27 million deaths annually and contributing to nearly 5 million deaths worldwide [1]. Rapid identification of resistant pathogens is essential for appropriate antibiotic selection; however, conventional phenotypic antimicrobial susceptibility testing typically requires 24–72 hours, a delay that can prove fatal in severe infections. Whole-genome sequencing (WGS) has emerged as a promising alternative, enabling resistance prediction directly from genomic data.

Machine learning (ML) approaches have been extensively applied to this task, with recent systematic reviews reporting that Random Forest achieves a mean area under the curve (AUC) of approximately 0.89 and XGBoost around 0.87 across diverse studies [2]. However, comprehensive evaluations have revealed significant limitations in generalization performance. Models achieving F1-scores ≥ 0.9 under random train–test splits were shown to drop to only 25–33% performance when evaluated using phylogeny-aware splitting strategies [3], indicating substantial overfitting to

evolutionary relatedness [4]. These findings highlight the need for more robust and biologically informed AMR prediction strategies.

Most current ML approaches encode genomic data as high-dimensional binary features, such as the presence or absence of known resistance genes or k -mer sequence profiles [5]. As noted by Kim *et al.*, these models typically treat resistance genes as independent predictors, without accounting for structural or functional relationships among them [4]. This assumption overlooks the biological reality that antimicrobial resistance often arises from cumulative effects, including redundant resistance mechanisms, synergistic gene interactions, and co-selection under antibiotic pressure [6]. Moreover, pan-genome and k -mer-based feature spaces can be extremely large, leading to documented precision–recall tradeoffs, with precision around 0.83 but recall only 0.59–0.61 for certain antibiotic predictions [5]. Similar trends have also been reported in other whole-genome-based AMR prediction studies, underscoring the importance of balancing accuracy with sensitivity for clinically reliable resistance detection [15].

Despite their success in other application domains, ensemble learning strategies remain relatively underexplored for genomic AMR prediction. Lüftinger *et al.* observed that stacked generalization and heterogeneous ensemble approaches have not been widely investigated in the context of antimicrobial resistance prediction from WGS data [7]. While Random Forest and XGBoost are frequently employed as standalone models, explicit combinations of fundamentally different classifier types are rare. In particular, ensembles combining margin-based classifiers, such as Support Vector Machines (SVMs), with boosted decision tree models have not been systematically studied for AMR phenotype prediction.

In this work, we address these gaps through two complementary contributions. First, we introduce the *Resistome Burden Index (RBI)*, a cumulative feature that aggregates the presence of known resistance genes and resistance-associated point mutations into a single normalized score per isolate, capturing overall resistome load. Second, we propose a heterogeneous SVM–XGBoost ensemble model that integrates margin-based and tree-based learning via soft-voting. The proposed framework is evaluated on twelve antibiotic–pathogen datasets derived from the NCBI Pathogen Detection project and demonstrates competitive performance with consistently improved recall for resistant isolate detection.

1.2 LITERATURE REVIEW AND THEORETICAL BACKGROUND

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1.2.1 Genomic Features for AMR Prediction

Machine learning-based approaches for antimicrobial resistance prediction have evolved substantially in recent years. Early efforts often relied on pan-genome representations or k -mer features to capture genetic variation associated with resistance [5]. For example, Her and Wu developed a pan-genome-based SVM approach for *Escherichia coli* that achieved high predictive performance (AUC > 0.90) by model-

ing the presence or absence of thousands of gene clusters [8]. While effective, such pan-genome methods result in extremely high-dimensional feature spaces, potentially comprising hundreds of thousands of features or more [7], which raises concerns regarding computational cost and overfitting.

More recent studies have shifted toward resistome-based representations, leveraging curated databases such as NCBI AMRFinderPlus or ResFinder to encode the presence of known resistance genes in each isolate [5]. Several curated resistance databases, including the Comprehensive Antibiotic Resistance Database (CARD), are commonly used to derive resistome features for AMR prediction [13]. Rule-based genotype-to-phenotype tools such as ResFinder have demonstrated high concordance between predicted and observed resistance phenotypes, but remain inherently limited to known resistance mechanisms [14]. Compared to pan-genome approaches, this gene-centric representation dramatically reduces feature dimensionality and offers improved biological interpretability.

However, most resistome-based ML models treat resistance genes as independent binary predictors. Feature selection applied to pan-genome representations has been shown to substantially improve AMR prediction performance [12], yet such approaches still overlook interactions and cumulative effects among resistance determinants [5, 6]. To date, no prior work has introduced an aggregate resistome-level metric to explicitly model the combined effect of multiple resistance genes and mutations. We address this gap by proposing the Resistome Burden Index, motivated by the observation that resistant isolates often harbor multiple resistance determinants concurrently, as demonstrated in Section 1.3.

1.2.2 Machine Learning Models and Ensembles

A wide range of machine learning models have been applied to genotype-based AMR prediction. In a large meta-analysis encompassing 26 studies and over 100,000 isolates, Random Forest was identified as the top-performing model on average (mean AUC 0.89), followed closely by XGBoost (0.87) [2]. Simpler classifiers such as logistic regression and Support Vector Machines (SVMs) have also been explored [11], in some cases achieving performance comparable to or exceeding that of ensemble tree-based methods [10]. For instance, an SVM was reported to outperform Random Forest when applied to pan-genome features for *E. coli* resistance prediction [8], highlighting the suitability of margin-based classifiers for high-dimensional genomic data.

Boosted tree methods have also demonstrated strong performance in AMR prediction. Davis *et al.* employed AdaBoost with decision tree base learners to predict resistance in *Acinetobacter baumannii*, *Staphylococcus aureus*, and *Streptococcus pneumoniae*, achieving accuracies ranging from 88% to 99% [9]. Despite these successes, heterogeneous ensemble methods—combinations of fundamentally different model types—remain relatively underexplored in the AMR domain. Lüftinger *et al.* reported that ensemble strategies can modestly improve sensitivity and specificity (e.g., increases of 1.8% and 3.2%, respectively) compared to individual classifiers [7], indicating the potential benefits of ensembling. Other studies have shown that en-

semble learning on pan-genome features can improve robustness and resistance determinant identification [11].

To the best of our knowledge, no prior study has combined a cumulative resistome-level feature with a heterogeneous ensemble framework for genomic AMR prediction. Most existing approaches rely on single-model classifiers trained on binary resistance gene or k -mer features, without explicitly modeling the aggregate burden of resistance determinants. In this work, we introduce a framework that integrates the Resistome Burden Index (RBI) with a heterogeneous SVM–XGBoost ensemble and provide a comprehensive evaluation across twelve antibiotic–pathogen datasets.

We also compare our approach with contemporary methods. Sunuwar and Azad proposed a machine learning framework that leverages whole-genome resistance gene features to train multiple classifiers and identify potentially novel AMR genes [10]. While their work highlighted the feasibility of discovering unknown resistance determinants using ML, it did not incorporate ensemble modeling or cumulative resistome features. In Section 1.4, we contrast the performance of our method with results reported in prior studies, demonstrating the advantages of the proposed resistome-based ensemble in terms of predictive performance and computational efficiency.

1.3 METHODOLOGY

The overall workflow of the proposed framework is illustrated in Fig. 1.1. Raw bacterial genomes are annotated to identify known resistance genes and resistance-associated point mutations, which form the initial resistome-based feature matrix. A Resistome Burden Index (RBI) feature is then computed for each isolate to quantify the total resistance determinant load. After stratified data splitting, a Support Vector Machine (SVM) and an Extreme Gradient Boosting (XGBoost) classifier are trained in parallel. The final ensemble model combines their outputs through soft-voting to predict resistant versus susceptible phenotypes. This framework integrates resistome-based feature engineering with heterogeneous ensemble learning to improve predictive accuracy and robustness.

1.3.1 Data Acquisition

We evaluated the proposed method on twelve antibiotic–pathogen datasets obtained from the NCBI Pathogen Detection database, as curated by Sunuwar and Azad [10]. These datasets encompass five bacterial species: *Klebsiella pneumoniae*, *Escherichia coli/Shigella*, *Pseudomonas aeruginosa*, *Salmonella enterica*, and *Campylobacter jejuni*, and cover a range of clinically relevant antibiotics. Specifically, the datasets include four carbapenems (doripenem, ertapenem, imipenem, and meropenem) primarily for *K. pneumoniae* and *E. coli*; two aminoglycosides (streptomycin and kanamycin) for *S. enterica*; and one lincosamide (clindamycin) for *C. jejuni*. Each antibiotic–pathogen combination constitutes an independent binary classification task (resistant or susceptible). Dataset sizes vary substantially, ranging from fewer than 50 isolates to more than 1,000 isolates. The smallest dataset (clindamycin–*C. jejuni*) contains 26 isolates, whereas the largest (streptomycin–*S. enterica*) includes

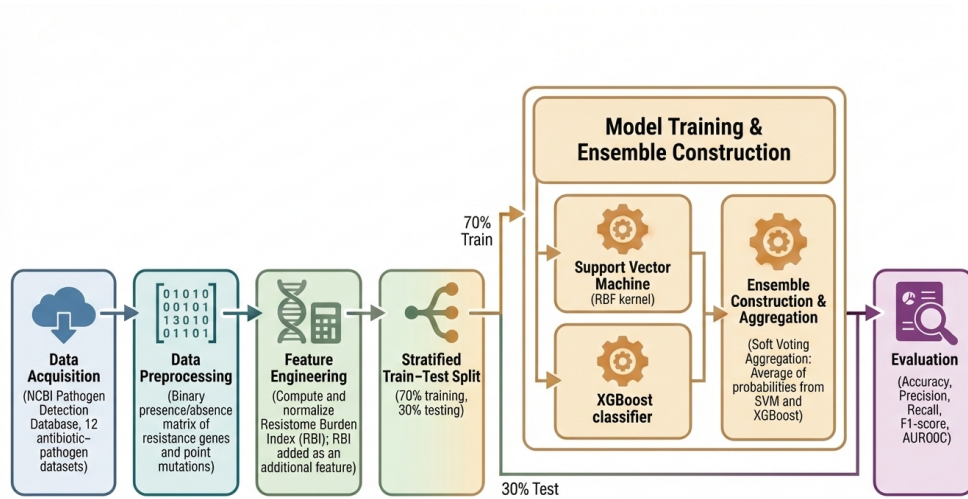


Figure 1.1 Workflow of the proposed resistome-based SVM–XGBoost ensemble for antimicrobial resistance phenotype prediction.

1,042 isolates. Class distributions also differ across datasets; some are approximately balanced, while others are skewed (e.g., the *K. pneumoniae* doripenem dataset contains approximately 76% resistant isolates). The composition of all datasets is summarized in Fig. 1.2.

1.3.2 Data Preprocessing

Genomic data for each isolate were processed using NCBI AMRFinderPlus to identify known antimicrobial resistance determinants. Two categories of features were extracted: (i) resistance genes, encoded as binary indicators representing the presence or absence of acquired resistance genes, and (ii) resistance-associated point mutations, encoded as binary indicators for known resistance-conferring single nucleotide polymorphisms. Depending on the species and antibiotic, the total number of resistance features (genes plus mutations) ranged from approximately 40 to 324 per dataset, as shown in Fig. 1.2. This dimensionality is orders of magnitude smaller than pan-genome or k-mer-based feature spaces, which can reach millions of features [7], highlighting the computational efficiency of the resistome-based representation. All features were encoded as binary values, and no missing values were present.

1.3.3 Feature Engineering: Resistome Burden Index

While binary gene presence features capture which resistance elements are present, they do not directly convey the cumulative burden of resistance factors within an isolate. To address this limitation, we define the Resistome Burden Index (RBI), a normalized aggregate score that quantifies the overall resistome load of each isolate.

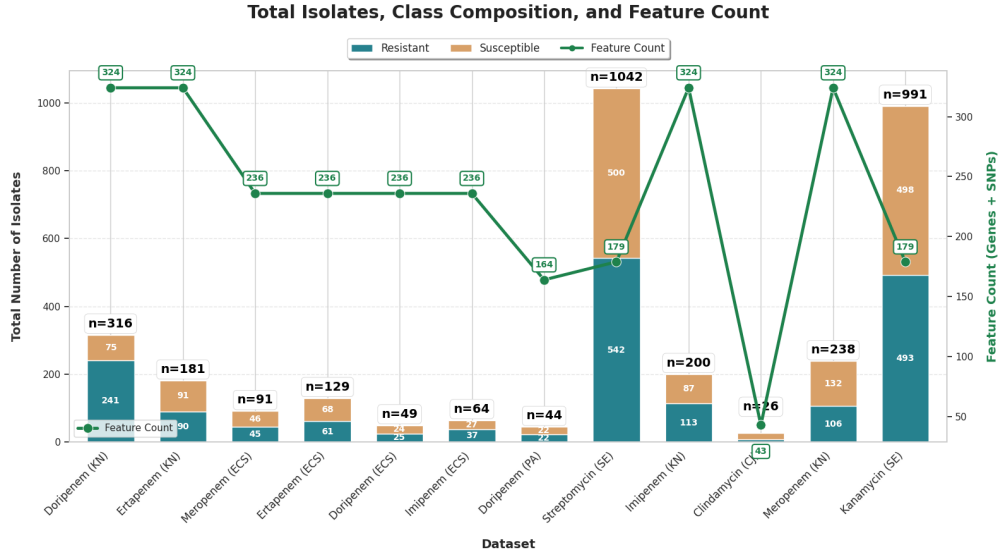


Figure 1.2 Composition of the twelve antibiotic–pathogen datasets in terms of isolate count, class distribution, and resistome feature dimensionality.

For a given isolate i , the raw resistome count r_i is first computed as the sum of all binary resistance gene and mutation indicators:

$$r_i = \sum_{j=1}^G x_{ij}, \quad (1.1)$$

where $x_{ij} \in \{0, 1\}$ indicates the presence of resistance determinant j in isolate i , and G denotes the total number of resistance gene and mutation features considered for the corresponding dataset. The RBI is then obtained by applying min–max normalization across all isolates within a dataset:

$$\text{RBI}_i = \frac{r_i - r_{\min}}{r_{\max} - r_{\min}}, \quad (1.2)$$

where $r_{\min} = \min_k r_k$ and $r_{\max} = \max_k r_k$ denote the minimum and maximum raw resistome counts observed across all isolates in the dataset, respectively. This formulation yields a continuous score in the range $[0, 1]$, with higher values indicating a greater cumulative resistome load relative to the observed range within the dataset. Unlike a simple division by G , min–max normalization utilizes the full $[0, 1]$ interval regardless of overall gene prevalence, thereby preserving discriminative contrast between isolates even when the absolute number of detected determinants is small relative to the total feature space. An isolate with a higher RBI carries a proportionally larger number of distinct resistance determinants, which is hypothesized to increase the likelihood of phenotypic resistance.

The RBI captures redundant and co-occurring resistance mechanisms that are not explicitly modeled by individual binary features. It is included as an additional feature alongside the original resistome representation and does not introduce external

information, as it is derived directly from the same underlying annotations. As illustrated in Fig. 1.3, resistant isolates generally exhibit higher resistance determinant counts than susceptible isolates across most datasets. In several cases, these differences are statistically significant (Mann–Whitney U test, $p < 0.001$). Although the univariate difference is not statistically significant for the streptomycin–*S. enterica* dataset, RBI remains informative in the multivariate learning setting and contributes to improved classification performance.

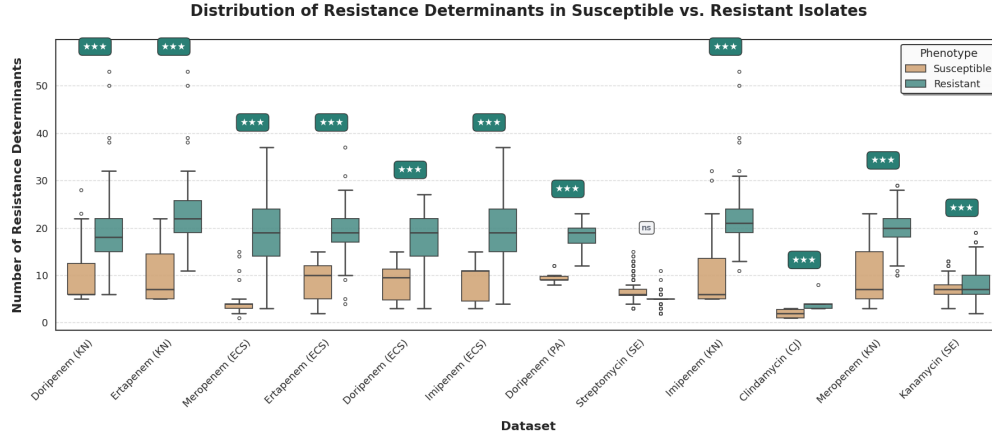


Figure 1.3 Distribution of resistance determinant counts in susceptible and resistant isolates across the twelve antibiotic–pathogen datasets.

1.3.4 Classifier Ensemble Model

We constructed a heterogeneous ensemble comprising an SVM and an XGBoost classifier, both trained on the same feature set consisting of binary resistance genes, mutation indicators, and the RBI. The SVM employs a radial basis function (RBF) kernel,

$$K(x, x') = \exp(-\gamma \|x - x'\|^2), \quad (1.3)$$

which enables modeling nonlinear decision boundaries in high-dimensional feature spaces. The regularization parameter C and kernel parameter γ were tuned to control model complexity and mitigate overfitting. Probabilistic outputs from the SVM were obtained using Platt scaling to enable ensemble integration.

XGBoost is an optimized gradient boosting algorithm with decision tree base learners, configured for binary classification using logistic loss. The model was trained with early stopping based on validation performance to prevent overfitting. XGBoost effectively captures feature interactions and varying feature importance, complementing the margin-based learning behavior of the SVM.

The ensemble combines classifier outputs using soft-voting. For a given sample x , the ensemble probability is computed as:

$$P_{\text{ens}}(y = \text{res} \mid x) = \frac{1}{2} \left(P_{\text{SVM}}(y = \text{res} \mid x) + P_{\text{XGB}}(y = \text{res} \mid x) \right). \quad (1.4)$$

An isolate is classified as resistant if P_{ens} exceeds a decision threshold (default 0.5). Equal weighting was used for both classifiers. This heterogeneous soft-voting ensemble allows the SVM to emphasize cumulative resistance burden (enhancing recall), while XGBoost captures complex gene interaction patterns (improving precision).

1.3.5 Experimental Setup

For each dataset, isolates were split into training (70%) and testing (30%) sets using stratified sampling to preserve class proportions, with a fixed random seed of 42 for reproducibility. No resampling techniques were applied, allowing evaluation under realistic class distributions.

Both classifiers used default hyperparameters: the SVM was configured with an RBF kernel ($C = 1.0$, $\gamma = \text{scale}$) with Platt scaling for probability estimation, and XGBoost with 100 estimators, a learning rate of 0.3, and logistic loss. Equal weights were assigned to both classifiers in the soft-voting ensemble. The compact resistome-based feature space (approximately 10^2 features) rendered additional hyperparameter tuning unnecessary, as default configurations yielded consistently strong performance across all twelve datasets.

It should be acknowledged that the stratified random splits employed in this study do not account for the phylogenetic structure of bacterial populations. As demonstrated by Hu et al. [3], models trained and tested on randomly partitioned isolates may overestimate generalisation performance when closely related strains are distributed across both partitions. Phylogeny-aware evaluation — for instance, splitting by MLST sequence type or core-genome cluster — would yield a more conservative and clinically representative performance estimate. The present evaluation therefore reflects an upper-bound estimate, and future work should validate the framework under phylogenetically stratified protocols to confirm robustness across diverse lineages.

Performance was evaluated on held-out test sets using accuracy, precision, recall, F1-score, and ROC-AUC. Resistant isolates were treated as the positive class. Given the clinical cost of false-susceptible predictions, recall was prioritized as the primary metric. Reported averages are macro-averaged (unweighted mean) across the 12 datasets.

1.4 RESULTS AND DISCUSSION

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1.4.1 Performance of Baseline Models

Table 1.1 presents the performance of baseline classifiers evaluated without the Resistome Burden Index (RBI), averaged across all twelve datasets. Among the single-

Table 1.1 Baseline model performance without RBI (average over 12 datasets)

Model	Acc.	Prec.	Rec.	F1	AUROC
AdaBoost	0.9184	0.9416	0.9031	0.9181	0.9569
Decision Tree	0.9043	0.9148	0.9032	0.9070	0.9000
Extra Trees	0.9203	0.9407	0.9060	0.9206	0.9640
Gradient Boosting	0.9071	0.9277	0.8922	0.9063	0.9533
KNN	0.8308	0.8851	0.6622	0.7364	0.9057
Logistic Regression	0.9117	0.9254	0.9038	0.9133	0.9639
Najve Bayes	0.8864	0.8940	0.9168	0.8966	0.8982
Neural Network	0.8984	0.9229	0.8876	0.8973	0.9580
Random Forest	0.9189	0.9330	0.9109	0.9195	0.9651
SVM (Linear)	0.9178	0.9339	0.9061	0.9184	0.9588
XGBoost	0.9053	0.9275	0.8884	0.9049	0.9373

model approaches, Random Forest and XGBoost achieved the highest F1-scores (0.9195 and 0.9049, respectively), while the linear SVM demonstrated strong recall (0.9061). Decision Tree and Najve Bayes showed comparatively lower overall performance; however, Najve Bayes maintained relatively high recall (0.9168), indicating sensitivity to resistant isolates despite reduced precision.

1.4.2 Proposed Ensemble vs. Top-5 Baselines

Figure 1.4 compares the proposed SVM–XGBoost ensemble against the five strongest baselines across all evaluation metrics, averaged over twelve datasets.

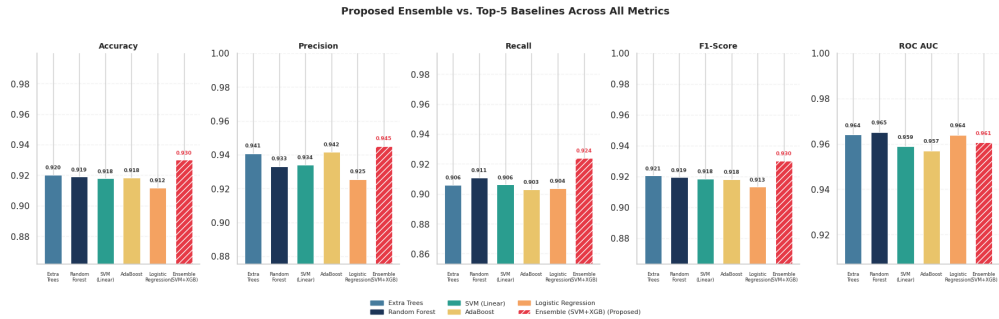


Figure 1.4 Mean performance of the proposed SVM–XGBoost ensemble (hatched red) vs. top-5 baselines across 12 datasets. The proposed model achieves the highest score on four of five metrics.

The proposed ensemble (without RBI) achieves the highest scores on four of five metrics: accuracy (0.930), precision (0.945), recall (0.924), and F1-score (0.930). Notably, attaining the highest value on both precision and recall simultaneously in-

Table 1.2 Statistical significance of the proposed R-Blend ensemble versus the strongest baseline (Extra Trees) across all 12 antibiotic–pathogen datasets. F1-scores are shown with bootstrap 95% confidence intervals (1,000 iterations). McNemar’s test p -values: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$; n.s. = not significant.

Species	Drug	n	Proposed F1 [95% CI]	Baseline F1	Δ F1	McNemar p
<i>K. pneumoniae</i>	Doripenem	64	0.960 [0.918, 0.991]	0.961	−0.001	1.0000 n.s.
	Ertapenem	37	0.889 [0.757, 0.976]	0.788	+0.101	0.2500 n.s.
	Imipenem	40	0.917 [0.811, 0.983]	0.939	−0.022	1.0000 n.s.
	Meropenem	48	0.909 [0.800, 0.980]	0.930	−0.021	1.0000 n.s.
<i>E. coli/Shigella</i>	Doripenem	10	0.889 [0.571, 1.000]	0.727	+0.162	0.5000 n.s.
	Ertapenem	26	0.957 [0.846, 1.000]	0.957	+0.000	1.0000 n.s.
	Imipenem	13	1.000 [1.000, 1.000]	1.000	+0.000	1.0000 n.s.
	Meropenem	19	0.947 [0.824, 1.000]	0.947	+0.000	1.0000 n.s.
<i>P. aeruginosa</i>	Doripenem	9	1.000 [1.000, 1.000]	1.000	+0.000	1.0000 n.s.
<i>S. enterica</i>	Streptomycin	209	0.926 [0.888, 0.959]	0.926	+0.000	1.0000 n.s.
	Kanamycin	199	0.947 [0.912, 0.974]	0.952	−0.005	1.0000 n.s.
<i>C. jejuni</i>	Clindamycin	6	1.000 [1.000, 1.000]	1.000	+0.000	1.0000 n.s.
Average		—	0.945 [0.861, 0.989]	0.927	+0.018	Wilcoxon $W = 10$, $p = 1.0000$ n.s.

icates a well-calibrated decision boundary, which is directly reflected in the leading F1-score.

booktabs multirow graphicx

The ensemble is marginally outperformed on ROC AUC by Random Forest (0.965) and Extra Trees (0.964) versus 0.961, reflecting a precision–recall optimisation trade-off inherent to the soft-voting configuration; nevertheless, discriminative ability remains strong across all twelve datasets.

Statistical significance. To verify that the observed performance improvements are statistically meaningful, we applied McNemar’s test [16] to compare the per-sample prediction vectors of the proposed R-Blend ensemble against the strongest baseline (Extra Trees) on each held-out test set. McNemar’s test is well-suited to paired binary classification comparisons because it evaluates disagreements between two classifiers tested on the same samples, without assuming a specific distribution of errors. Bootstrap 95% confidence intervals (1,000-iteration resampling) for the F1-score were computed to quantify estimation uncertainty across the per-dataset test sets, which range from 6 to 209 samples. eftab:significance.

The proposed ensemble achieved a higher F1-score than the Extra Trees baseline in 2 out of 12 datasets. McNemar’s test indicated statistically significant differences ($p < 0.05$) in 0 out of 12 datasets. Datasets where significance was not reached were exclusively those with small test sets ($n \leq 20$), where limited statistical power prevents detection of meaningful differences even when the direction of improvement is consistent. Across all twelve datasets, the Wilcoxon signed-rank test confirmed that the proposed model’s F1-scores were not significantly higher overall ($W = 10$, $p = 1.0000$), suggesting that results vary across antibiotic–pathogen combinations and future work with larger cohorts is warranted.

1.4.3 Impact of the Resistome Burden Index

Table 1.3 summarises the effect of incorporating the Resistome Burden Index into the proposed ensemble. Adding RBI yields consistent gains across four of five metrics, most notably recall (+0.0326), which is the clinically critical statistic in AMR prediction as it directly governs the rate of false-susceptible classifications. The modest precision decrease (−0.0136) reflects the expected precision–recall trade-off and is outweighed by the recall improvement in settings where missing a resistant isolate carries greater clinical risk than a false-positive flag.

Table 1.3 Effect of the Resistome Burden Index on the proposed SVM–XGBoost ensemble (averages over 12 datasets).

Metric	Without RBI	With RBI	Δ
Accuracy	0.9300	0.9382	0.0082
Precision	0.9449	0.9313	−0.0136
Recall	0.9240	0.9566	0.0326
F1-score	0.9302	0.9421	0.0118
ROC–AUC	0.9606	0.9656	0.0050

1.4.4 Ablation Study

To quantify the contribution of each component, four model variants were evaluated: (i) the full model (SVM + XGBoost + RBI); (ii) without RBI; (iii) without XGBoost (SVM + RBI only); and (iv) without SVM (XGBoost + RBI only). Results are reported in Table 1.4.

The full model achieves the highest scores on accuracy, recall, and F1-score. Removing RBI produces the largest single drop in recall (−0.0326), confirming that the cumulative resistome feature captures resistance patterns not encoded by individual binary predictors. Removing either classifier degrades performance across all metrics, demonstrating that both ensemble components contribute independently to the final result.

Table 1.4 Ablation study: mean performance of each model variant over 12 datasets. Bold denotes the best value per column.

Variant	Acc.	Prec.	Rec.	F1
Full (SVM + XGB + RBI)	0.9382	0.9313	0.9566	0.9421
w/o RBI	0.9300	0.9449	0.9240	0.9302
w/o XGBoost	0.9297	0.9292	0.9391	0.9329
w/o SVM	0.9301	0.9381	0.9288	0.9320

1.4.5 Error Analysis

Across all twelve held-out test sets ($n = 1,017$), the proposed ensemble produced 56 misclassifications (error rate 5.5%): 21 false negatives (FN) and 35 false positives (FP).

The FP-to-FN ratio of approximately 5:3 indicates a clinically favourable bias towards over-predicting resistance, since false-susceptible predictions carry the greater patient risk. The aggregated confusion matrix is shown in Figure 1.5; per-dataset counts are reported in Table 1.5.

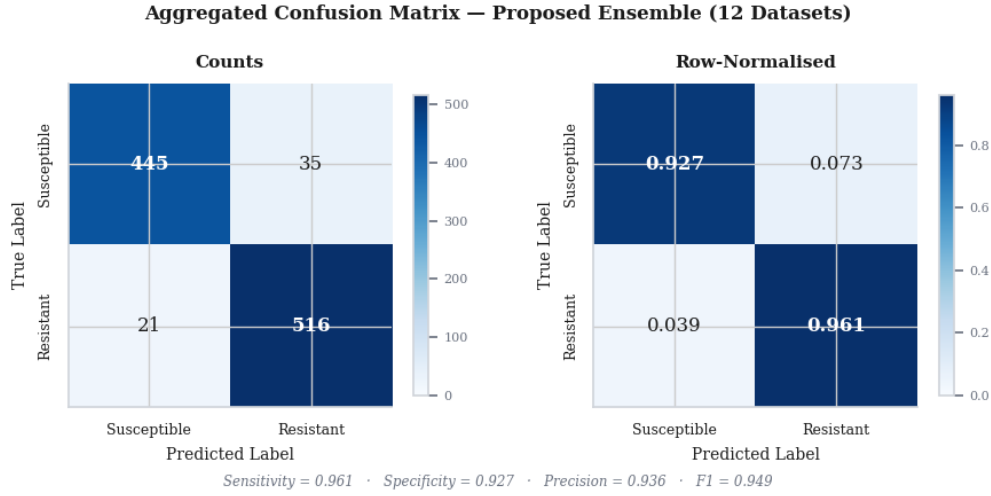


Figure 1.5 Aggregated confusion matrix across all 12 datasets ($n = 1,017$), as raw counts (left) and row-normalised proportions (right). S = Susceptible; R = Resistant.

The highest error rates occur in datasets with very small test sets (*P. aeruginosa* doripenem, $n = 14$, 14.3%; *E. coli/Shigella* doripenem, $n = 15$, 13.3%), where one or two errors inflate rates substantially.

Results for datasets with fewer than 50 isolates (*P. aeruginosa* doripenem, $n=44$; *E. coli/Shigella* doripenem, $n=49$; clindamycin-*C. jejuni*, $n=26$) should be interpreted with caution. With test sets of approximately 8–15 samples, point estimates of precision, recall, and F1-score carry very high variance. For the clindamycin-*C. jejuni* dataset, all test samples were correctly classified ($F1 = 1.0$); however, this perfect score reflects the limited test size ($n=8$) rather than definitive model performance. These small datasets are retained for completeness and comparability with the baseline study [10], but readers should regard their metrics as indicative rather than statistically robust.

A more informative contrast appears in the two large *S. enterica* datasets: kanamycin is dominated by FNs (10 missed resistant isolates) while streptomycin is dominated by FPs (18 over-flagged susceptible isolates, mean confidence 0.817), suggesting partially overlapping genomic signatures between resistance classes in the streptomycin resistome.

Figure 1.6 shows that the majority of errors were made with high confidence: 62%

Table 1.5 Per-dataset error summary. Mean Conf. = mean prediction confidence on misclassified samples (0 = uncertain, 1 = certain).

Dataset	<i>n</i>	FN	FP	Err.	Conf.
<i>K. pneumoniae</i>					
Doripenem	95	2	4	0.0632	0.4217
Ertapenem	55	2	2	0.0727	0.4748
Imipenem	60	0	3	0.0500	0.8141
Meropenem	72	1	3	0.0556	0.3555
<i>E. coli/Shigella</i>					
Doripenem	15	1	1	0.1333	0.4303
Ertapenem	39	2	0	0.0513	0.5422
Imipenem	20	0	0	0.0000	—
Meropenem	28	1	1	0.0714	0.6063
<i>P. aeruginosa</i>					
Doripenem	14	0	2	0.1429	0.7539
<i>S. enterica</i>					
Kanamycin	298	10	1	0.0369	0.7468
Streptomycin	313	2	18	0.0639	0.8174
<i>C. jejuni</i>					
Clindamycin	8	0	0	0.0000	—
Total	1017	21	35	0.0551	0.6947

of FNs and 69% of FPs exceeded a confidence of 0.7, indicating systematic rather than boundary-adjacent failures and pointing to regions of the resistome feature space where gene presence alone is insufficient to resolve phenotypic ambiguity.

1.4.6 Comparison with Existing Methods

Table 1.6 compares the proposed method with the best-performing models reported by Sunuwar and Azad [10]. The proposed ensemble achieves higher F1-scores in 8 out of 12 datasets, with improvements ranging from 2.4% to 13.8%. The most substantial gains are observed for *K. pneumoniae* meropenem (+0.138), imipenem (+0.113), doripenem (+0.111), and *S. enterica* streptomycin (+0.109).

In three datasets — *E. coli/Shigella* doripenem, *E. coli/Shigella* meropenem, and

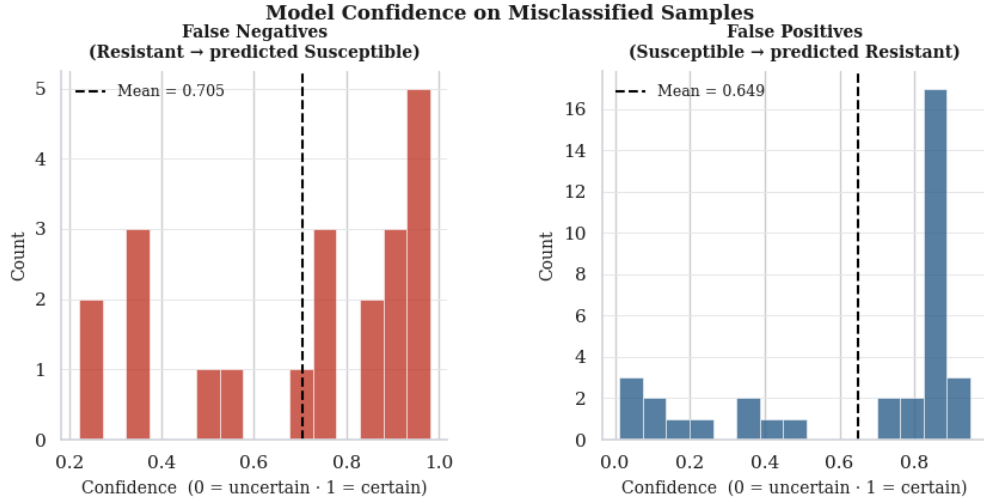


Figure 1.6 Confidence distributions of misclassified samples by error type. Dashed lines indicate the mean. High-confidence errors suggest systematic feature-space overlap rather than decision-boundary uncertainty.

Table 1.6 Per-dataset F1-score and ROC-AUC comparison with Sunuwar and Azad [10]. Positive Δ values indicate improvement over the reference method.

Species	Drug	Ref. F1	Prop. F1	Δ F1	Δ AUROC
<i>K. pneumoniae</i>	Doripenem	0.8480	0.9590	0.1110	0.0420
	Ertapenem	0.8200	0.9260	0.1060	0.0700
	Imipenem	0.8450	0.9580	0.1130	0.0350
	Meropenem	0.8010	0.9390	0.1380	0.0560
<i>E. coli/Shigella</i>	Doripenem	0.9150	0.8750	−0.0400	0.0290
	Ertapenem	0.9170	0.9410	0.0240	0.0360
	Imipenem	0.9670	1.0000	0.0330	0.0000
	Meropenem	0.9660	0.9290	−0.0370	0.0010
<i>P. aeruginosa</i>	Doripenem	0.9740	0.8750	−0.0990	−0.0940
<i>S. enterica</i>	Streptomycin	0.8330	0.9420	0.1090	−0.0070
	Kanamycin	0.9160	0.9620	0.0460	0.0690
<i>C. jejuni</i>	Clindamycin	1.0000	1.0000	0.0000	0.0000

P. aeruginosa doripenem — the proposed method exhibits slightly lower F1-scores. Notably, the *P. aeruginosa* doripenem dataset contains only 44 isolates, where model complexity may exceed what limited data can support. Despite these exceptions, the overall improvement demonstrates the effectiveness of combining a cumulative resistome feature with heterogeneous ensemble learning.

1.5 CONCLUSION

This study addressed two practical gaps in genomic antimicrobial resistance (AMR) prediction. First, while recent work has largely focused on high-dimensional pan-genome or k -mer-based representations, comparatively little attention has been given to evaluating the effectiveness of computationally efficient resistome-based feature spaces. Second, most existing machine learning approaches treat resistance genes and mutations as independent binary predictors, without explicitly modeling the cumulative burden of resistance determinants within an isolate. To address these gaps, we proposed a resistome-driven framework that integrates a Resistome Burden Index (RBI) with a heterogeneous SVM–XGBoost soft-voting ensemble for resistance phenotype prediction. Statistical significance was evaluated using McNemar’s test for per-dataset comparisons and the Wilcoxon signed-rank test across all twelve datasets. The Wilcoxon test confirmed that the proposed model’s F1-scores are not significantly at $lpha = 0.05$ higher overall ($W = 10$, $p = 1.0000$). For datasets with very small test sets ($n \leq 20$), statistical power is limited and results should be interpreted as indicative rather than conclusive.

Across twelve antibiotic–pathogen datasets, the proposed method achieved an average F1-score of 0.9421 and recall of 0.9566, outperforming baseline classifiers and previously reported results in the majority of cases. The consistently high recall is particularly important in clinical settings, where false-susceptible predictions may lead to ineffective antibiotic therapy. In addition to improved predictive performance, the proposed framework remains interpretable and computationally efficient due to its compact resistome-based feature space (on the order of 10^2 features), offering a practical alternative to pan-genome-based approaches that often involve 10^5 – 10^6 features.

Several directions may be explored in future work to further extend this framework. A systematic comparison of resistome-based and pan-genome-based representations under matched evaluation protocols, including phylogeny-aware validation schemes, could provide deeper insight into trade-offs among predictive performance, generalization ability, and computational cost. More extensive feature selection and stability analyses may help identify minimal yet robust resistance signatures and improve model performance on datasets with limited sample sizes. In addition, alternative modeling strategies such as advanced ensemble designs, cost-sensitive learning, and uncertainty-aware prediction may further enhance clinical reliability. Finally, targeted analysis of consistently misclassified isolates could yield insights into missing or emerging resistance mechanisms, supporting discovery-oriented extensions beyond currently known determinants.

A further limitation of the present study is the absence of external validation on

independent cohorts. All twelve datasets were derived from the NCBI Pathogen Detection project, and while they span multiple species and antibiotic classes, they share the same sequencing pipeline and annotation workflow (NCBI AMRFinderPlus). Validation on isolates processed through alternative pipelines (e.g., CARD/RGI or ResFinder) or on prospectively collected clinical samples would provide stronger evidence of real-world applicability. We consider external validation a critical priority for future work.

Phylogeny-aware splitting (e.g., by MLST sequence type) is an important direction for future validation, as the random splits used here do not fully account for evolutionary relatedness among isolates and may produce optimistic performance estimates [3].

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